Existence and Uniqueness for the Weak Poisson Problem

An End-to-End Meta-Math Walkthrough

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Abstract

The gradient form becomes an inner product once a Poincaré estimate controls the missing L^2 part. Riesz representation then constructs the weak solution directly. The report presents the target theorem, declared imports, a dependency graph, five-component node contracts, a continuous proof, a worked example, and a hypothesis-repair audit. It is an authored mathematical exposition, not a deployed-agent transcript or a formal proof-assistant certificate.

PDE-02 / Partial Differential Equations / Report v1

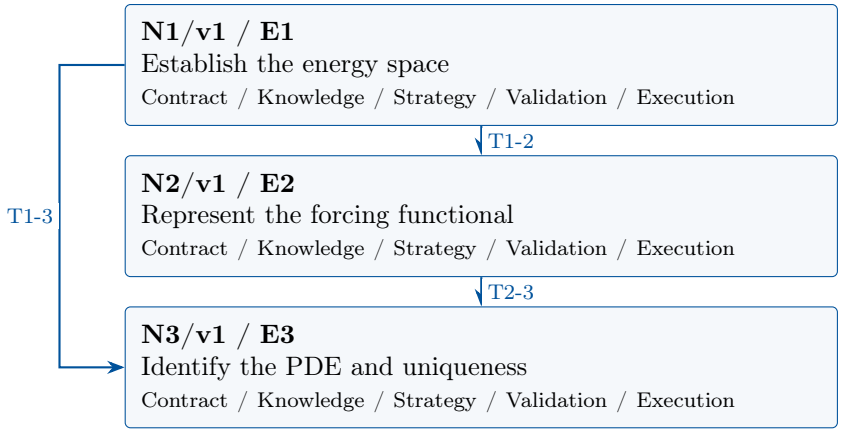
I. TARGET AND IMPORT CONTRACT

Theorem I.1. *Let $\Omega \subset \mathbb{R}^d$ be nonempty, bounded, and open, and let real $f \in L^2(\Omega)$. Prove there is a unique $u \in H_0^1(\Omega)$ such that $\int_\Omega \nabla u \cdot \nabla v = \int_\Omega fv$ for every $v \in H_0^1(\Omega)$. If Ω lies inside a cube of side L, prove $\|\nabla u\|_2 \le L\|f\|_2$. This is a weak solution of $-\Delta u = f$ with zero boundary condition encoded by H_0^1.*

A. Foundations and permitted imports

Define $H^1(\Omega)$ as the L^2 functions whose distributional first derivatives lie in L^2, with norm squared $\|u\|_2^2 + \|\nabla u\|_2^2$. Import completeness of this Sobolev space; H_0^1 is the closure of $C_c^\infty(\Omega)$ in that norm. Import the real Hilbert-space Riesz theorem FA-03: a bounded linear functional on a Hilbert space is inner product with a unique vector, of equal norm. Use Cauchy–Schwarz, Fubini on smooth functions, and the one-dimensional fundamental theorem. The needed Poincaré bound is proved here.

II. NODE INTERACTION WORKFLOW



Each box is one mathematical agent containing all five functional components. Blue arrows carry the versioned exports listed below. The layout gives a reading order; only drawn arrows assert dependencies. Diagnostic repair is separate from the proof DAG.

A. Versioned evidence-transfer ledger

T1-2: N1/E1/v1 to N2. $\|v\|_2 \le L\|\nabla v\|_2$ on H_0^1, and $a(u,v) = \int \nabla u \cdot \nabla v$ is a complete inner product there.

Receiving activity. N2.1

Compatibility. E1/v1 and $\ell(v) = \int_\Omega fv$. Check: The functional is defined on equivalence classes and bounded in the gradient norm, not merely in the L^2 norm.

T1-3: N1/E1/v1 to N3. $\|v\|_2 \le L\|\nabla v\|_2$ on H_0^1, and $a(u,v) = \int \nabla u \cdot \nabla v$ is a complete inner product there.

Receiving activity. N3.2

Compatibility. E2/v1 and the definition of distributional Laplacian. Check: The conclusion is weak existence; no C^2 regularity or pointwise boundary trace on a rough domain is inferred.

T2-3: N2/E2/v1 to N3. There exists $u \in H_0^1$ with $a(u,v) = \ell(v)$ for all v, and $\|\nabla u\|_2 \le L\|f\|_2$.

Receiving activity. N3.1

Compatibility. E2/v1 and the definition of distributional Laplacian. Check: The conclusion is weak existence; no C^2 regularity or pointwise boundary trace on a rough domain is inferred.

III. AGENT ROLES AND FIVE-COMPONENT ACTIVITIES

A. N1/v1: Establish the energy space

Object and role. Integrate along one coordinate to prove Poincaré, then use density and equivalent norms. **Local obligation:** $\|v\|_2 \le L\|\nabla v\|_2$ on H_0^1, and $a(u,v) = \int \nabla u \cdot \nabla v$ is a complete inner product there.

Inputs. A cube Q of side L containing Ω and smooth compactly supported test functions extended by zero to Q.

C — Contract

Publish E1/v1 only after establishing this obligation: $\|v\|_2 \le L\|\nabla v\|_2$ on H_0^1, and $a(u,v) = \int \nabla u \cdot \nabla v$ is a complete inner product there.

K — Knowledge

A cube Q of side L containing Ω and smooth compactly supported test functions extended by zero to Q. Retain the stated assumptions and import versions.

S — Strategy

Integrate along one coordinate to prove Poincaré, then use density and equivalent norms.

V — Validation

Zero extension is used first only for compactly supported smooth functions; the general inequality follows by the defining closure.

E — Execution

Carry out activities N1.1–N1.2 below, retaining each calculation and construction as evidence.

Numbered mathematical activities

N1.1 After translating the cube, take $Q = (0,L)^d$. For $v \in C_c^\infty(\Omega)$ extended by zero, $v(x_1,x') = \int_0^{x_1} \partial_1 v(s,x')\,ds$. Cauchy–Schwarz gives $|v(x_1,x')|^2 \le L\int_0^L |\partial_1 v(s,x')|^2 ds$. Integrate over x_1 and x' to obtain $\|v\|_2^2 \le L^2\|\partial_1 v\|_2^2 \le L^2\|\nabla v\|_2^2$.

N1.2 Approximate any $v \in H_0^1$ by such smooth functions in the H^1 norm. Passing to the norm limit preserves the inequality. Therefore $\|\nabla v\|_2 = 0$ forces $v = 0$, and the gradient norm is equivalent to the complete H^1 norm on its closed subspace H_0^1. A gradient-Cauchy sequence is H^1-Cauchy by the inequality and converges in both norms. Thus a is a Hilbert inner product.

Export E1/v1. $\|v\|_2 \le L\|\nabla v\|_2$ on H_0^1, and $a(u,v) = \int \nabla u \cdot \nabla v$ is a complete inner product there.

Acceptance gate. Zero extension is used first only for compactly supported smooth functions; the general inequality follows by the defining closure.

Targeted repair. If the gradient seminorm is called a norm without justification, prove Poincaré first; constants are not automatically excluded in other Sobolev spaces. Withhold this export and recheck its descendants after a versioned correction.

B. N2/v1: Represent the forcing functional

Object and role. Bound the forcing in the energy norm and apply the declared Hilbert representation theorem. **Local obligation:** There exists $u \in H_0^1$ with $a(u,v) = \ell(v)$ for all v, and $\|\nabla u\|_2 \le L\|f\|_2$.

Inputs. E1/v1 and $\ell(v) = \int_\Omega fv$. Incoming proof exports: E1/v1 at N2.1.

C — Contract

Publish E2/v1 only after establishing this obligation: There exists $u \in H_0^1$ with $a(u,v) = \ell(v)$ for all v, and $\|\nabla u\|_2 \le L\|f\|_2$.

K — Knowledge

E1/v1 and $\ell(v) = \int_\Omega fv$. Incoming proof exports: E1/v1 at N2.1. Retain the stated assumptions and import versions.

S — Strategy

Bound the forcing in the energy norm and apply the declared Hilbert representation theorem.

V — Validation

The functional is defined on equivalence classes and bounded in the gradient norm, not merely in the L^2 norm.

E — Execution

Carry out activities N2.1–N2.2 below, retaining each calculation and construction as evidence.

Numbered mathematical activities

N2.1 Cauchy–Schwarz and E1 give $|\ell(v)| \le \|f\|_2\|v\|_2 \le L\|f\|_2\|\nabla v\|_2$. Thus ℓ is a bounded linear functional on the real Hilbert space with inner product a.

N2.2 Apply FA-03 to obtain a unique representing vector u with $\ell(v) = a(v,u) = a(u,v)$, since the form is symmetric. The norm equality in Riesz yields $\|\nabla u\|_2 = \|\ell\| \le L\|f\|_2$. This is exactly the asserted variational equation.

Export E2/v1. There exists $u \in H_0^1$ with $a(u,v) = \ell(v)$ for all v, and $\|\nabla u\|_2 \le L\|f\|_2$.

Acceptance gate. The functional is defined on equivalence classes and bounded in the gradient norm, not merely in the L^2 norm.

Targeted repair. If the functional is represented using the usual H^1 inner product, change to a; otherwise an unwanted zeroth-order term appears. Withhold this export and recheck its descendants after a versioned correction.

C. N3/v1: Identify the PDE and uniqueness

Object and role. Test against compactly supported smooth functions and against the difference of two proposed solutions. **Local obligation:** The vector u solves $-\Delta u = f$ in distributions and is the unique stated weak solution.

Inputs. E2/v1 and the definition of distributional Laplacian. Incoming proof exports: E1/v1 at N3.2, E2/v1 at N3.1.

C — Contract

Publish E3/v1 only after establishing this obligation: The vector u solves $-\Delta u = f$ in distributions and is the unique stated weak solution.

K — Knowledge

E2/v1 and the definition of distributional Laplacian. Incoming proof exports: E1/v1 at N3.2, E2/v1 at N3.1. Retain the stated assumptions and import versions.

S — Strategy

Test against compactly supported smooth functions and against the difference of two proposed solutions.

V — Validation

The conclusion is weak existence; no C^2 regularity or pointwise boundary trace on a rough domain is inferred.

E — Execution

Carry out activities N3.1–N3.2 below, retaining each calculation and construction as evidence.

Numbered mathematical activities

N3.1 For $\varphi \in C_c^\infty(\Omega) \subset H_0^1$, the distribution $-\Delta u$ acts by $\int \nabla u \cdot \nabla \varphi$. The variational identity equals $\int f\varphi$, proving the distributional equation. Membership in H_0^1 is the specified homogeneous boundary condition.

N3.2 If u_1, u_2 solve the equation, their difference w satisfies $a(w, v) = 0$ for every $v \in H_0^1$. Choosing $v = w$ gives $\|\nabla w\|_2^2 = 0$, and E1 gives $w = 0$. This independently verifies uniqueness and closes the problem.

Export E3/v1. The vector u solves $-\Delta u = f$ in distributions and is the unique stated weak solution.

Acceptance gate. The conclusion is weak existence; no C^2 regularity or pointwise boundary trace on a rough domain is inferred.

Targeted repair. If a weak solution is promoted to a classical one, remove that unsupported conclusion or supply a separate regularity theorem with its required domain assumptions. Withhold this export and recheck its descendants after a versioned correction.

IV. CONTINUOUS MATHEMATICAL PROOF

A. N1 — Establish the energy space

1. **N1.1.** After translating the cube, take $Q = (0, L)^d$. For $v \in C_c^\infty(\Omega)$ extended by zero, $v(x_1, x') = \int_0^{x_1} \partial_1 v(s, x')\, ds$. Cauchy–Schwarz gives $|v(x_1, x')|^2 \le L \int_0^L |\partial_1 v(s, x')|^2 ds$. Integrate over x_1 and x' to obtain $\|v\|_2^2 \le L^2 \|\partial_1 v\|_2^2 \le L^2 \|\nabla v\|_2^2$.

2. **N1.2.** Approximate any $v \in H_0^1$ by such smooth functions in the H^1 norm. Passing to the norm limit preserves the inequality. Therefore $\|\nabla v\|_2 = 0$ forces $v = 0$, and the gradient norm is equivalent to the complete H^1 norm on its closed subspace H_0^1. A gradient-Cauchy sequence is H^1-Cauchy by the inequality and converges in both norms. Thus a is a Hilbert inner product.

B. N2 — Represent the forcing functional

1. **N2.1.** Cauchy–Schwarz and E1 give $|\ell(v)| \le \|f\|_2 \|v\|_2 \le L \|f\|_2 \|\nabla v\|_2$. Thus ℓ is a bounded linear functional on the real Hilbert space with inner product a.

2. **N2.2.** Apply FA-03 to obtain a unique representing vector u with $\ell(v) = a(v, u) = a(u, v)$, since the form is symmetric. The norm equality in Riesz yields $\|\nabla u\|_2 = \|\ell\| \le L \|f\|_2$. This is exactly the asserted variational equation.

C. N3 — Identify the PDE and uniqueness

1. **N3.1.** For $\varphi \in C_c^\infty(\Omega) \subset H_0^1$, the distribution $-\Delta u$ acts by $\int \nabla u \cdot \nabla \varphi$. The variational identity equals $\int f \varphi$, proving the distributional equation. Membership in H_0^1 is the specified homogeneous boundary condition.

2. **N3.2.** If u_1, u_2 solve the equation, their difference w satisfies $a(w, v) = 0$ for every $v \in H_0^1$. Choosing $v = w$ gives $\|\nabla w\|_2^2 = 0$, and E1 gives $w = 0$. This independently verifies uniqueness and closes the problem.

□

V. WORKED ORCHESTRATION

On $\Omega = (0, 1)$ with $f = 1$, let $u(x) = x(1 - x)/2$. It belongs to H_0^1: smooth cutoffs near the endpoints converge in H^1, since u vanishes linearly there and the cutoff-derivative errors are supported on intervals of vanishing length. It satisfies $-u'' = 1$. Integration by parts against smooth compactly supported v gives $\int u'v' = \int v$, and density extends this to all H_0^1. Its energy norm is $\|u'\|_2 = 1/\sqrt{12} \le 1 = L\|f\|_2$.

A. Boundary and hypothesis audit

On $H^1(\Omega)$ instead of H^1_0, a nonzero constant has zero gradient, so the energy form is not an inner product. Testing the same proposed equation with $v = 1$ would require $\int f = 0$, a compatibility condition absent from the Dirichlet theorem. Changing the space changes the problem.

VI. ILLUSTRATIVE FAILURE AND REPAIR

The following is an authored diagnostic scenario. Changed assumptions define a new contract; they do not retroactively prove the original statement.

A. Rejected argument, diagnosis, and correction

Rejected N1 export: the gradient norm is complete and positive on all H^1. **Validation:** nonzero constants have norm zero in that seminorm. **Repair:** retain H^1_0 and the explicit Poincaré estimate before applying Riesz. The original weak Dirichlet target is unchanged. Recheck the forcing bound, representation, and uniqueness downstream.

B. Publication and reuse

The corrected export retains its explicit assumptions. A hypothetical revision receives a new version, and each consuming activity is rechecked; this scenario is not a recorded runtime event. **Next direction:** Develop Lax–Milgram for nonsymmetric coercive forms, or establish additional regularity under smoother domain and forcing assumptions.

Provenance. This authored proof is complete relative to its declared imports. Reuse each export with its assumptions, proof, and version. The original Markdown remains available for comparison.

Further reading. Leon Simon, Lectures on PDE, for elliptic, weak, and parabolic methods. Classical calculus and the explicitly stated Hilbert-space imports are used below; the displayed PDE arguments are developed in full.